

Self-attention is a mechanism used in Transformer model that allows each word (or token) in a sequence to attend to every other word in the same sequence including itself. It helps the model understand relationship b/w words ~~regardless~~ regardless of their position.

→ attention in transformers is a technique that lets each word in a sequence attend (look at) other words in the same sequence and assign different weights to them based on how important they are for understanding or generating the context.

Self attention v/s attention in transformer

Feature Self-attention v/s attention in transformer

Self attention v/s Traditional attention

Aspect	Traditional attention	Self attention in Transformer
Scope	aligns two Sequence - typically input and output (e.g. encoder-decoder in translation)	operations within a single Sequence (e.g. a sentence) allowing elements to attend to other
Computation of weight	weights are based on similarity b/w decoder state and encoder hidden state	Uses queries, keys & values from the input to compute similarity b/w all element
Application	works sequentially by calculating attention at each decoding step	works in parallel for the entire sequence, allowing faster processing
Types of dependencies	capture cross sequence dependencies (e.g. input output alignment)	capture intra sequence dependencies (e.g. word to word relationship b/w a sentence)
Efficiency	slower and less scalable for long sequence due to step by step computation	more efficient & scalable thanks to parallel computation

word Embedding : ~~Vector~~ word embedding are the mathematical vector representations of words. they capture the semantic meaning of word so that similar word have similar vector.

~~word embeddings are numerical v.~~
or word embeddings are numerical vector representation of words they capture the meaning of words in a way that similar words are close together in vector space.

Famous word embedding technique used using model like word2vec, Glove, or BERT.

e.g "King" $\rightarrow [0.9, 0.2, -0.5, \dots]$
"Queen" $\rightarrow [0.91, 0.21, -0.49, \dots]$

these vectors are close together meaning they are semantically related.

Why use word Embedding?

1) They allow model to understand semantic meaning of word

2) they convert text into a format that machine learning model can use.

Vector Database:- vector databases are specialized database that store and retrieve vector embeddings (from text, image, audio, etc)

or

A vector database stores & retrieves vector representation (like word embeddings, image embeddings etc) efficiently. it is used for similarity search (finding things that are close in meaning or appearances).

Key concept:

- traditional database store structured data (tables, row) but struggle with unstructured data like image or free text.
- vector database bridge the semantic gap by storing data as vector embeddings
- Similar data is stored closer in vector space, making it easy to perform semantic search.

Common vector DB's FAISS, Pinecone, etc
what they store:-

- word and sentence embeddings
- Image embedding
- audio/video content in vector form

Use Cases:

- 1) Semantic Search
- 2) Recommendation system,
- 3) Medical image analysis
- 4) NLP with Retrieval-Augmented Generation (RAG)